Fiscal Year:	FY 2005	**Task Last Updated:**	FY 11/01/2005
PI Name:	Bateman, Ted A. Ph.D.		
Project Title:	Examination of Anti-Resorptive and Anabolic Treatments/Stimuli on Unloading Induced Osteoporosis		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Bone Loss Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Bone Fracture**:Risk of Bone Fracture due to Spaceflight-induced Changes to Bone (2) **Osteo**:Risk Of Early Onset Osteoporosis Due To Spaceflight		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Chapel Hill	**State:**	NC
Zip Code:	27599	**Congressional District:**	4
Comments:	Previous affiliation was Clemson University; PI moved to UNC in fall 2010.		
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	07/01/2004	**End Date:**	06/30/2008
No. of Post Docs:	0	**No. of PhD Degrees:**	5
No. of PhD Candidates:	5	**No. of Master' Degrees:**	7
No. of Master's Candidates:	7	**No. of Bachelor's Degrees:**	9
No. of Bachelor's Candidates:	9	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Simske, Steven (University of Colorado) Norrdin, Robert (Colorado State University)		
Grant/Contract No.:	NCC 9-58-BL00403		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Studies examining the calcium balance during extended spaceflight on the Russian Space Station Mir suggest that 1) both bone resorption and formation are altered to adversely affect bone density, and 2) the time required to return bone mass to pre-flight levels may be considerably longer than the period of microgravity exposure. Countermeasures to prevent the loss of bone mass, and therapies to more quickly replace bone post-flight, are desirable to not only prevent fractures and kidney stones during and shortly after long-duration missions, but to also prevent microgravity-associated bone loss from exacerbating naturally occurring osteoporosis. Given the severity of bone loss, the long-term health concerns, and the numerous biomedical problems associated with spaceflight, every effort should be made to develop countermeasures that require minimal intervention. In the first year of this grant we began the development of a treatment regimen that would prevent in-flight bone loss and accelerate the recovery of bone health post-flight. Both bone mass and turnover rates are important for evaluating successful recovery. The significance of these factors will be illustrated by first demonstrating that low doses of anti-resorptive agents can prevent in-flight loss of bone mass and allow bone turnover to recover more quickly. Fundamental to this objective was the determination of the minimum, efficacious therapeutic doses of anti-resorptives with studies employing mice administered either osteoprotegerin (OPG) or zoledronate. Following this, approximately 100 µgkg-1 for OPG and 45 µgkg-1 for zoledronate were determined to be the minimum, efficacious comparable doses. For the second phase of the project (to be initiated in the second year of funding), OPG will be combined with anabolic agents. The disuse osteoporosis model of hind limb suspension, followed by a period of loaded recovery, will be used. Low-dose OPG will be administered during unloading and the anabolic therapeutic will be examined during loaded recovery. The anabolic agents being examined are also designed to develop a minimally invasive regimen: exercise, ultrasound and parathyroid hormone (PTH), in this order, are the preferred methods of returning bone health. The second year of the grant will focus on PTH as the anabolic agent, and then examine the other stimuli (exercise and ultrasound). PTH is the only clinically approved anabolic therapy for osteoporosis and may be necessary in cases where an astronaut has lost an unusually large amount of bone or when turnover does not return to pre-flight levels. In addition to the primary project outlined, two synergistic projects have progressed during the first year of the grant. 1) A clinical examination of sequential bisphosphonate and PTH treatment via a retrospective study was performed, and 2) an examination of the effects of space radiation on bone mass was initiated. Progress on these projects will be discussed through the report and will be referred to as 1) the clinical project and 2) the radiation project.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The human skeletal system consists of a dynamic balance between solid, calcified bone and free Ca2+ in the serum. Two processes, resorption and synthesis, consistently turnover bone and contribute to so-called “bone remodeling”. These processes are completed by osteoclasts and osteoblasts, respectively. An imbalance in this system, such as in post-menopausal women or during spaceflight, can result in a shift favoring bone resorption. Taken to its extreme this imbalance results in hypercalcemia, which is often an indicator of a disorder of bone metabolism called osteoporosis, characterized by low bone density, bone mass and an increase risk of fracture. Although the mechanisms by which bone loss occur naturally, such as in post-menopausal woman, and during spaceflight are different, research in the area of spaceflight induced bone loss has profound implications for the pathogenesis of osteoporosis on earth, and in particular the development of effective therapies to prevent and reverse bone loss. Indeed, studies on the international space station reveal that bone loss occurs at a rate much faster than of post-menopausal woman. With over 1.5 million fractures occurring per year due to the proliferation of an aging population and the expansion of space exploration to unprecedented new lengths, the need for the development of a variety of advanced, effective therapies has never been greater. Osteoprotegerin (OPG) is a soluble protein that blocks osteoclast formation and therefore bone resorption by acting as a decoy receptor to prevent the binding of RANKL to the RANK receptor. OPG is an important regulator of bone mass and shows great potential as a therapeutic treatment for osteoporosis in post-menopausal women (currently in FDA Phase III clinical trials as AMG 162, in development by the biotechnology company Amgen). Bisphosphonates are non-hydrolizable analogues of pyrophosphate that inhibit bone resorption through their high affinity for bone minerals and their inhibitory effects on osteoclasts. Alendronate, risedronate and ibandronate are bisphosphonates approved for the treatment of osteoporosis, and zoledronate is approved to treat hypercalcemia and is in clinical trial to treat osteoporosis. Current research is looking at the capacity for these drugs to prevent hypercalcemia and bone loss in animal models of disuse. Estrogen normally acts on osteoblasts in order to promote OPG production and the maintenance of normal bone structure; however, menopause is characterized by a sharp decrease in circulating estrogen levels and results in a decrease in circulating OPG. Without this inhibitory effect osteoclastogenesis increases dramatically. During spaceflight osteoclast activation through skeletal unloading also leads to bone loss. The present study looks to find an effective dosing regiment of OPG that is able to prevent hypercalcemia and prevent bone loss without having the same effects on turn over or incorporation into bone as bisphosphonates. Such findings will surely be beneficial to both ground-based and spaceflight application. Previously the undesirable and harmful side effects of radiation therapy were seen as secondary to the primary goal of eradicating malignancies. With the development of better and more targeted radiation therapy, resulting in longer and more productive survival, the prevention of the increased fracture rates in patients receiving radiation therapy is critical. Indeed this has an important clinical relationship to the issue of low dose spaceflight radiation exposure, which leads to bone loss much in the way that previous studies have shown irradiated animals lose significant bone mass. Skeletal unloading and radiation-induced bone loss are the primary factors contributing to spaceflight-induced osteoporosis. The implications for this work on earth are clear and represent the chance to more forward on the development of important clinic therapeutics for multiple cause
	The determination of the minimum, efficacious doses of OPG and zoledronate was obtained through a series of iterative studies, with the dose yielding maximal increases in trabecular density and/or connectivity in one study serving as the medium dose for the next study. With each iteration the bracketing dose range on the medium dose was narrowed. In the first study with 0.3, 3 and 30 ug/kg ZOL and 1, 10, and 100 ug/kg OPG, doses of 30 ug/kg ZOL and 100 ug/kg produced maximal density response. Thus doses of 6, 30, 300 ug/kg ZOL and 100, 500, 2500 ug/kg OPG were selected for the next study. Maximal density response to 30 ug/kg ZOL and 100 ug/kg again narrowed the range for the next iteration. The final study used 10, 30 and 45 ug/kg ZOL, while the OPG range was significantly narrowed to 100, 250 and 500 ug/kg. Finally, doses of 45 ug/kg ZOL and 100 ug/kg OPG were selected as the minimum, efficacious doses based on a combination of trabecular density and connectivity data. In addition to determining the dose that results in maximal increases in trabecular density, a hypercalcemia model is also being employed. This is a short-term model, where hypercalcemia is induced via continuous administration of PTH (as opposed intermittent administration that is used as an anabolic therapy). The goal is to determine the minimum dose of OPG and zoledronate that will prevent

Task Progress:	hypercalcemia. These data will be combined with the doses determined for the maximal trabecular density model. Because the literature is more complete for rats rather than mice, it was necessary to perform two preliminary studies in order to develop an appropriate hypercalcemia model. Indeed, mice appear to require a higher dose of PTH to induce hypercalcemia than rats do (i.e. 80 ug/kg/day rather than 40 ug/kg/day). For the clinical project, where a chart review is examining the efficacy of PTH as an anabolic therapy with and without prior treatment with bisphosphonates, the preliminary data suggest an important effect of prior bisphosphonate treatment. A study of 105 patients shows trends that indicate prior bisphosphonate treatment may inhibit or delay the anabolic effect of PTH. For the radiation project, where skeletal tissue from mice and rats exposed to multiple sources of radiation (gamma, proton, carbon and iron) were examined, we have identified that simulated space radiation causes significant declines in bone mass. Two sets of bones were sent to us from Loma Linda University Medical Center. Hindlimbs were obtained from rats exposed to 0, 1, 2 and 4 Gy of iron radiation and from mice exposed to 2 Gy of gamma, proton, carbon and iron radiation. Both studies significant declines in bone mass and/or trabecular connectivity were observed for all doses/groups. Serum markers for bone formation and resorption from the iron exposed rats have helped us identify that the bone loss is likely caused by osteoclastic bone resorption. Media coverage has consisted of front page coverage on the Anderson (SC) Independent newspaper (July 14th, 2005). Coverage related to return to flight (STS-114) and our previous spaceflight experiments and work at KSC. I was down at KSC for a talk to the Spaceflight and Life Sciences Training Program and brought a local high school teacher who is working in my lab for the summer with me.
Bibliography Type:	Description: (Last Updated: 11/12/2020)
Articles in Peer-reviewed Journals	Bateman TA. "Molecular therapies for disuse osteoporosis." Gravit Space Biol Bull. 2004 Jun;17(2):83-9. , Jun-2004
Awards	Hamilton, SA, Gridley, DS, Pecaut, MJ, Bateman, TA. "3rd Place Overall Poster Award for: Hamilton, SA, Gridley, DS, Pecaut, MJ, Bateman, TA The Effects of Radiation on Murine Bone Loss. International Academy of Astronautics, Human in Space Symposium, Graz, Austria, May 2005." May-2005
Awards	Hamilton, SA, Gridley, DS, Pecaut, MJ, Bateman, TA. "Student Travel Award for: Hamilton, SA, Gridley, DS, Pecaut, MJ, Bateman, TA Osteoblast and Osteoclast Response to Radiation Induced Bone Loss. Radiation Research Society Annual Meeting, Denver, CO, October 2005." Oct-2005
Presentation	Bateman, TA "Space biomedicine: a tool for engineering and biology education." N/A Apr-2005
Presentation	Bateman, TA "Nerve damage induced osteolysis: a new interdisciplinary biomedical disorder." N/A Sep-2005
Presentation	Bateman, TA "Roadmap to mice on STS-108." N/A Jul-2004
Presentation	Bateman, TA, Hamilton, SA, Nelson, G "How do animal models inform the debate?" N/A Jun-2005
Presentation	Bateman, TA, Hamilton, SA, Nelson, G "Space radiation and bone loss." N/A Jul-2005
Presentation	Bateman, TA, Hamilton, SA, Nelson, G "The role of animal models in countermeasure development: bone loss in space." N/A Jul-2005
Presentation	Bateman, TA, Miesse, AM "Mice in Space!" N/A Oct-2004
Presentation	Hamilton, SA, Bateman, TA "Living and Working in Space" N/A Oct-2004
Presentation	Hamilton, SA, Miesse, AM, Mohr, KE, Howard, SH, Bandstra, ER, Bateman, TA "Microgravity and Biomedicine" N/A Dec-2004
Presentation	Miesse, AM, Hamilton, SA, Milstead, JR, Bandstra, ER, Mohr, KE, Bateman, TA "Spaceflight and the Skeletal System" N/A Jul-2004
Presentation	Miesse, AM, Willey, JS, Hamilton, SA, Bateman, TA "Fundamentals and Benefits of Animal Research" N/A Jul-2005
Presentation	Willey, JS, Hamilton, SA, Bandstra ER, Miesse, AM, Travis, N, Lau, A, Goodman, D, Bateman, TA "Osteoporosis Biomechanics Laboratory." N/A Jul-2005
Presentation	Yuan, YY, , Morony, S, Ferguson, VL, Simske, SJ, Lacey, DL, Warmington, KS, Geng, Z, Tan, HL, Shalhoub, V, Dunstan, CR, Kostenuik, PJ, Bateman, TA "OPG Prevents Relative Osteopenia and Deficits in Skeletal Strength in Mice During a 12 Day Spaceflight" N/A May-2005
Presentation	Yuan, YY, Lau, AG, Kostenuik,, PJ, Morony, S, Adamu, S, Asuncion, F, Bateman, TA "RANKL Causes Cortical and Cancellous Osteopenia and Reduces Bone Strength in Mice" N/A May-2005